Jane Street's Protocol Emulator ASIC Competition

Authors:
Rudy Magaña
Rogelio Correa
Connor Wilson

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Chapter 1

Challenge

Design an **open-source, general purpose** protocol emulator. More specifically, hardware communication protocols s.a.

- UART
- I2C
- SPI
- low-speed USB
- 10Mbit Ethernet
- JTAG, SWD, CAN, ...

1.1 What is a protocol emulator?

A protocol emulator is a small chip/block that mimics the behavior of a device or protocol. This is achieved with a tiny CPU with an instruction set designed for

- reading/writing pins
- counting cycles
- hit timing well enough to implement a real protocol in **firmware**.

1.2 Tips

- Recommended work flow:
  - Run synthesis
  - Check mapped cell area
  - leave room for clk-tree buffers and routing
  - Run full place-and-route
  - Check timing
- SRAM is area efficient over flip-flops for instr memory
- Start by getting a UART transmitter out of a pin. Then make it programmable...

Chapter 2

Deliverables/Requirements

2.1 Constraints

2.1.1 Jane Street Rules

Process : IHP's 130nm CMOS5L through TinyTapeout.

- 6×4 is 24 tiles of $200\mu m \times 150\mu m$. $\sim 1K$ logic cells
- TODO : set `tiles` in `info.yaml` to 6×4

2.1.2 Tiny Tapeout

What the heck does TT need?? YAML?

Chapter 3

Design

3.1 Protocols

3.1.1 UART

3.1.2 SPI

3.1.3 I2C

3.2 RTL...

Chapter 4

Verification

Chapter 5

Issues

Chapter 6

Conclusion

Chapter 7

Appendix

7.1 Code

7.1.1 SystemVerilog

7.1.2 others...

Chapter 8

References

- Jane Street Website
- TinyTapeout's CMOS5L Verilog Template